Scalar Characters and Contact-Height Estimates

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Abstract

For the normalized quaternionic representation used in the five-contact construction, the square of the nebentypus is the inverse square of the scalar Dirichlet character. Choosing the local scalar characters of minimal two-power order bounds the nebentypus by the largest two-part of $p-1$ among the relevant primes. This gives a sharper modular degree bound, an explicit cubic logarithmic contact estimate with constant less than 2^{330670}, and a finer estimate sensitive to congruence restrictions on the contact primes. Separately, evaluation of differentials at a primitive sixth root of unity removes the horizontal contribution to the discriminant estimate for modular Jacobians. The arithmetic realization remains an explicitly stated input.

1 Scalar characters of minimal order

For a positive squarefree integer R divisible by 6, define

$$U(R) = \max\big(\{2\} \cup \{2^{v_2(p-1)} : p \mid R,\ p \text{ odd prime}\}\big), \qquad E(R) = \prod_{p \mid R} (1 + p^{-1}).$$

Here $v_2(n)$ is the exponent of 2 in the nonzero integer n. Both $U(R)$ and $E(R)$ are effectively computable.

Lemma 1.1. *Prescribe a subset of the finite places dividing R at which the local boundary of a Dirichlet character under the squaring sequence is to be nontrivial. The character used to realize these prescribed boundaries can be chosen tame at every odd prime, of conductor at most 4 at 2, and of order dividing $U(R)$. Moreover $U(R) \mid \varphi(R)$.*

Proof. At an odd prime p, put $p - 1 = 2^a u$ with u odd. If a nontrivial local boundary is prescribed, choose a generator g of $(\mathbb{Z}/p\mathbb{Z})^\times$ and set $\chi_p(g) = \zeta_{2^a}$, a primitive 2^a-th root of unity. Then

$$\chi_p(-1) = \zeta_{2^a}^{(p-1)/2} = -1.$$

Thus χ_p is odd and has the required boundary. The boundary criterion is the local reciprocity calculation used in the scalar lifting construction: the obstruction to taking a square root of a finite local character is its value at -1. If the boundary is trivial, use the trivial character. At 2, use the character modulo 4 when needed. The product of these characters has order dividing the least common multiple of their orders. All orders are powers of 2, so that least common multiple divides $U(R)$. The global boundary is the prescribed sign class whenever the local prescription comes from that class, by the same reciprocity and global-to-local argument as in the scalar lifting construction. Finally each $2^{v_2(p-1)}$ divides $\varphi(R) = \prod_{p \mid R,\ p>2}(p-1)$, and this product is even. The largest of these powers, together with 2, therefore divides $\varphi(R)$. □

2 The determinant identity

Let A be an abelian surface over a finite normal extension $L/\mathbb{Q}$. Suppose that compatible isogenies $\mu_\sigma : {}^\sigma A \to A$ have been chosen, and that the normalized semilinear operators

$$\rho_\ell(\sigma) = b(\sigma)^{-1} a_\sigma \mu_\sigma \circ \sigma, \qquad a_\sigma = (\deg \mu_\sigma)^{-1/4}, \qquad b^2 = \chi, \tag{1}$$

form a four-dimensional ℓ-adic representation. Coefficient fields are enlarged to contain these scalars, with the trivial Galois action on the added coefficients. Assume that the quaternionic action gives $\rho_\ell = r_\ell \oplus r_\ell$ and that modularity identifies r_ℓ with a weight-two newform representation, in the Tate-module convention

$$\det r_\ell = \varepsilon \chi_{\mathrm{cyc}}.$$

Here χ_{cyc} denotes the ℓ-adic cyclotomic character and ε is the nebentypus. These are precisely the normalized representations and conventions in the modular construction of [1, proof of `o8-height`, (LR2)–(LR5)].

Proposition 2.1. *The character $\varepsilon\chi$ has order at most two. Consequently, if χ is chosen as in Lemma 1.1, then*

$$\mathrm{ord}(\varepsilon) \mid U(R).$$

Proof. For an abelian surface, cup product and the trace isomorphism give the natural identification

$$\bigwedge^4 V_\ell(A) \simeq \mathbb{Q}_\ell(2).$$

An isogeny acts on this determinant line by its degree. This follows equally from its action on the top cohomology of a complex abelian surface and comparison; the exterior-algebra description of abelian cohomology is recalled in [3, Chapter I, Section 12]. Galois acts on the determinant line by χ_{cyc}^2. Consequently, even though the isogeny has a conjugate source,

$$\det(\mu_\sigma \circ \sigma) = \deg(\mu_\sigma)\chi_{\mathrm{cyc}}(\sigma)^2.$$

Taking determinants in (1), and using $a_\sigma^4 \deg(\mu_\sigma) = 1$, gives

$$\det \rho_\ell = b^{-4}\chi_{\mathrm{cyc}}^2 = \chi^{-2}\chi_{\mathrm{cyc}}^2.$$

On the other hand, $\det \rho_\ell = (\det r_\ell)^2 = \varepsilon^2 \chi_{\mathrm{cyc}}^2$. Cancelling the cyclotomic factor proves $\varepsilon^2 = \chi^{-2}$. Thus $\kappa = \varepsilon\chi$ is quadratic or trivial. The orders of both χ and κ divide $U(R)$, since $U(R)$ is even. Therefore the order of $\varepsilon = \kappa\chi^{-1}$ also divides $U(R)$. □

The quadratic twist of b used to remove unwanted ramification does not change this calculation: its fourth power is one, and its square leaves χ unchanged. The proposition bounds the nebentypus directly; it does not bound the newform coefficient-field degree.

3 A unit fibre for differential evaluation

We use the original stable Faltings height. Let $X_H = X_1(N)/H$, where $H \subset (\mathbb{Z}/N\mathbb{Z})^\times$ contains -1, and let J_H be its Jacobian of positive dimension g_H. Write

$$d_H = [X_H : \mathbb{P}^1_j] = N[(\mathbb{Z}/N\mathbb{Z})^\times : H] \prod_{p \mid N} (1 + p^{-1}).$$

Proposition 3.1. *One has*

$$h_{\mathrm{F}}(J_H) \le \frac{d_H}{2}(\log 6 + \log N + 2\log d_H) - \frac{g_H}{2}\log \pi.$$

Proof. Use $t = j/1728$ and let ζ be a primitive sixth root of unity. Choose a number field K over which the curve is semistable, the fibre $t = \zeta$ splits, and ζ is rational. Normalize $\mathbb{P}^1_{\mathcal{O}_K}$ in $K(X_H)$ and specialize its affine algebra at $t = \zeta$ to obtain an order $C \subset \mathcal{O}_K^{d_H}$. The integral evaluation argument of [2, Sections 1–3] maps the Hodge lattice into the trace-dual order. After projection to g_H coordinates on which evaluation is injective, the target's arithmetic degree is at most half the logarithmic discriminant of C. Injectivity holds because $d_H > 2g_H - 2$ for this Belyi map.

Both ζ and $1 - \zeta$ are algebraic units. The fibre therefore meets none of the horizontal branch divisors $0, 1, \infty$ at finite places. At primes greater than three, the torsion-different argument of [2, Lemma 4.2] bounds the normalized vertical contribution by $d_H \log N$. At 2 and 3, Lenstra's different estimate gives at most $d_H(\log 6 + 2 \log d_H)$, exactly as in [2, Section 4]. These arguments are unchanged after the extension to K; its degree cancels after normalization. Thus

$$\frac{\log \mathrm{N}\mathfrak{D}(C)}{[K : \mathbb{Q}]} \le d_H(\log 6 + \log N + 2 \log d_H).$$

At every complex embedding, the distance from the image of ζ to each of 0 and 1 is one. Every disk of radius $s < 1$ about that image is unramified, and its inverse image consists of disjoint analytic sheets. The mean-value inequality bounds evaluation of ω/dt by $1/(\sqrt{\pi} s)$ times the L^2 norm of ω. Letting s tend to one gives operator norm at most $1/\sqrt{\pi}$. Hadamard's inequality bounds the determinant norm by $\pi^{-g_H/2}$. The arithmetic degree inequality now gives the displayed result. □

4 The refined contact estimate

Put

$$B_* = \prod_{p \le 76176} p, \qquad C_0 = 210^{300}, \qquad \mathcal{D} = \{0, 1, 2401, -343/32\}.$$

For $a = u/v \in \mathbb{Q} \setminus \mathcal{D}$ in lowest terms with $v > 0$, let $M = M_5(a)$ be the product of primes greater than 76176 at which at least one of

$$v, \quad u, \quad u - v, \quad u - 2401v, \quad 32u + 343v$$

has positive odd valuation. Write $h(a) = \log \max(|u|, v)$.

The arithmetic premise in this section is the normalized modular construction of [1, proof of `o8-height`], including its geometric inequality $h(a) \le 2 \max(0, h_{\mathrm{F}}(A)) + 128$, its CM bound $h_{\mathrm{F}}(A) < 854R$, and in the non-CM case its surface factor on a modular Jacobian over L', its level bound $N \le C_0 R^2$, and $\mathrm{rad}(N) \mid R$, where $R = B_* M$. The scalar characters in that construction may be chosen as in Lemma 1.1; its conductor estimates use only tameness at odd primes and conductor at most 4 at 2. Its field construction gives $[L' : \mathbb{Q}] \le 1152 \, \mathrm{ord}(\chi)$. This retains the source's arithmetic premises and uses more information than a bare height inequality.

Theorem 4.1. *Under this premise, define*

$$C_3' = 19 C_0 B_*^3 \log B_*, \qquad C_2' = 19 C_0 B_*^2 E(B_*) \log B_*.$$

Then

$$h(a) \le C_3' M^3 (1 + \log M), \tag{2}$$

$$h(a) \le C_2' M^2 U(B_* M) E(M) (1 + \log M). \tag{3}$$

Moreover $C_3' < 2^{330670}$, *and hence*

$$h(a) < 2^{330671} M^{7/2}.$$

Proof. Let $H = \ker \varepsilon$. Proposition 2.1 gives

$$d_H \le C_0 R^2 U(R) E(R) =: \mathcal{T}. \tag{4}$$

Since $U(R) \mid \varphi(R)$,

$$U(R)E(R) \le \varphi(R)E(R) = R \prod_{p \mid R} (1 - p^{-2}) < R, \qquad \mathcal{T} < C_0 R^3.$$

Also $[L' : \mathbb{Q}] \le 1152 U(R) \le 1152 R$.

For completeness, the numerical height transfer can be made with the same estimates for both the refined and uniform bounds. Write $\ell = \log R$. The exact inequalities $C_0^2 < B_*$, $144 < B_*$ and $\ell \ge 1$ give

$$\log N \le \tfrac{5}{2}\ell, \qquad \log d_H \le \tfrac{7}{2}\ell, \qquad \log 12 \le \tfrac{1}{2}\ell.$$

The weaker version of Proposition 3.1 with $\log 12$ in place of $\log 6$ and with additive term $g_H \log(2/\sqrt{\pi})$ yields $h_{\mathrm{F}}(J_H) \le 6 d_H \ell$. Thus $H_J = \max(1, h_{\mathrm{F}}(J_H) + 3g_H/2) \le 8\mathcal{T}\ell$.

Rémond's subvariety inequality and the dimension-two isogeny bound, in the form proved in [2, Lemma 5.1], imply

$$h_{\mathrm{F}}(A) \le H_J + 8\left(32 \log 14 + \log d' + \log \max(1, \log d', H_J)\right), \qquad d' = [L' : \mathbb{Q}].$$

Here $\log d' < 9\ell$ and $\log(8\mathcal{T}\ell) \le 9\ell$, whereas $8\mathcal{T}\ell > \max(1, \log d')$. Using $\log 14 < 3$ gives

$$h_{\mathrm{F}}(A) \le 8\mathcal{T}\ell + 912\ell \le 9\mathcal{T}\ell.$$

The CM alternative is smaller than this bound as well. Since $\mathcal{T}\ell > 128$, the geometric comparison gives $h(a) \le 19\mathcal{T}\ell$. Substituting (4), using $E(R) = E(B_*)E(M)$ and $\log R \le (\log B_*)(1 + \log M)$, proves (3). The inequality $\mathcal{T} < C_0 R^3$ proves (2).

The exact checker described below verifies

$$19 C_0 B_*^3 \cdot 109445 < 2^{330670}, \qquad B_* < 2^{109445}.$$

Since $\log 2 < 1$, these inequalities imply $C_3' < 2^{330670}$. Finally $(1+u)e^{-u/2} < 2$ for $u \ge 0$, so the stated pure-power bound follows on taking $u = \log M$. □

Corollary 4.2. *Fix an integer $k \ge 1$. Suppose that every prime dividing $M_5(a)$ satisfies $v_2(p-1) \le k$. Then*

$$h(a) \le C_2' \max(65536, 2^k) M_5(a)^2 (1 + \log M_5(a))^2.$$

In particular, for every fixed $\eta > 0$ there is an effective constant $C_{k,\eta}$ such that $h(a) \le C_{k,\eta} M_5(a)^{2+\eta}$ on this restricted set.

Proof. The primes in B_* have largest two-part 65536, attained at 65537; this is certified by exact trial division. Thus $U(B_* M) \le \max(65536, 2^k)$. For squarefree M,

$$E(M) = \sum_{d \mid M} \frac{1}{d} \le \sum_{d=1}^{M} \frac{1}{d} \le 1 + \log M.$$

Apply (3). The remaining logarithmic factor is absorbed effectively by M^η, since $\sup_{u \ge 0} (1+u)^2 e^{-\eta u} < \infty$. □

For example, primes $p \equiv 17 \pmod{32}$ satisfy $v_2(p-1) = 4$. The corollary applies when *all* prime factors in the relevant contact radical obey that restriction. The freedom to choose such primes in a completeness argument does not impose the restriction on arbitrary rational solutions in a soundness argument. No such restriction on all contact primes is inferred here.

Proposition 4.3. *For $\beta(X) = 51X^2 + 2504$, the numbers $U(B_* M_5(\beta(X)))$, with $X \in \mathbb{Z}$, are unbounded. More precisely, for every $k \geq 1$ there are infinitely many primes $p > 76176$ with $v_2(p-1) \geq k$ for which infinitely many integers X satisfy $v_p(\beta(X)) = 1$.*

Proof. Set $L = 2^{k+3} \cdot 51 \cdot 2504$. Dirichlet's theorem on primes in arithmetic progressions [4, Section 1] supplies infinitely many primes $p \equiv 1 \pmod{L}$. Discard the finitely many primes at most 76176. For every odd prime q dividing $51 \cdot 2504$, quadratic reciprocity gives $(q/p) = (p/q) = 1$, because $p \equiv 1 \pmod{4q}$. Also $(-1/p) = (2/p) = 1$. Hence $-2504/51$ is a nonzero square modulo p. Choose x_0 with $51x_0^2 + 2504 \equiv 0 \pmod{p}$. Its derivative $102x_0$ is nonzero modulo p. Among the p residue classes $x_0 + pt \pmod{p^2}$, exactly one makes β vanish modulo p^2; any of the others has $v_p(\beta) = 1$. Each such residue class contains infinitely many integers. For these inputs the contact at zero contributes p to $M_5(\beta(X))$, so $U(B_* M_5(\beta(X))) \geq 2^{v_2(p-1)} \geq 2^k$. □

Thus the congruence condition in Corollary 4.2 cannot hold with one fixed k throughout this scalar family, even if the input is already restricted to the rational integers.

5 Exact verification and arithmetic scope

The program

`computation/contact_heights/check_nebentypus_height.py`

constructs B_* by a sieve and independently by trial division, checks $U(B_*) = 65536$, verifies the integer inequalities in Theorem 4.1, and tests the character-order identities in cyclic groups. It also checks the two equations for a primitive sixth root, $\zeta^2 - \zeta + 1 = 0$ and $\zeta(1-\zeta) = 1$. These finite checks verify the constants and elementary algebra. The determinant, duality, and arithmetic-realization assertions are the mathematical proofs and premises stated above, not conclusions of the computation.

References

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